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RobustLayoutLM: Leveraging Optimized Layout with Additional Modalities for Improved Document Understanding

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***Abstract*—Pre-training methods have become the mainstream in document understanding, involving self-supervised learning on large-scale unlabeled document to learn rich feature representations, followed by supervised fine-tuning on a smaller labeled document dataset. Recently emerged pre-training models for document understanding all use layout information as important features, however, their performance significantly suffers when the layout information is incorrect. In this paper, we introduce RobustLayoutLM, which utilizes the optimized layout information generated by our proposed Uniform XY Cut algorithm. And Visual-Graph Tagger is introduced to integrate image and graph information after the transformer feature fusion, minimizing the impact of erroneous layout information. Experimental results show that our RobustLayoutLM achieves competitive or better results on multiple standard VrDU benchmarks and outperforms previous methods in the face of incorrect layout information.**

I. Introduction

Visually-rich Document Understanding (VrDU) aims to automatically extract key information from image or PDF formatted documents. Such as extracting the issue date and tax amount from invoices; gathering the contract date and signatories from contract documents. Unlike the entity extraction task in the field of natural language processing, VrDU handles two-dimensional information, where the same text token can have different meanings depending on its spatial location in the document. As shown in Figure 1, the form filling date and authorization date contain identical text tokens, but have different meanings because of their different locations in the document.

For all VrDU methods, the two-dimensional layout information is critically important. However, in practical applications, the document layout information must be obtained by OCR tools, which can lead to instances of inaccuracies in layout information. We found that incorrect layout information severely affects the performance of multimodal VrDU models. As shown in Table I, changes in layout information lead to a significant decrease in the F1 scores of the text-layout models and image-text-layout models on the VrDU benchmark, with the incorrect layout information affecting the feature expression of other modalities.

In this paper, we introduce RobustLayoutLM, a VrDU model with greater robustness, capable of maintaining good performance under various layout conditions. In RobustLayoutLM, we present a novel Uniform XY Cut algorithm to reorder the misaligned bounding boxes and enhance the layout information while preserving the original shape of text blocks. Unlike previous methods that generate image embeddings before feeding features into the transformer, we choose to extract image features after the transformer feature fusion, based on document token bounding boxes regenerated by the Uniform XY Cut algorithm, thus preventing errors in corresponding image features caused by incorrect layout information.

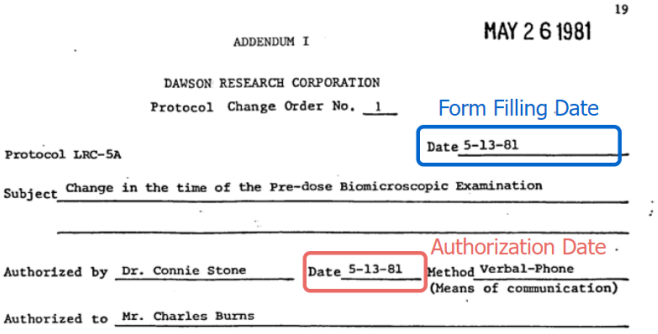


Fig. 1: An example from FUNSD dataset. The text in the two boxes is the same.

Method	Changes in F1 score	Pre-training Modality
$\text{LayoutLM}_{\text{BASE}}$	78.66 → 66.62	Text + Layout
$\text{BROS}_{\text{BASE}}$	83.05 → 61.04	Text + Layout
$\text{LiLT}_{\text{BASE}}$	88.41 → 68.93	Text + Layout
$\text{LayoutLMv2}_{\text{BASE}}$	82.76 → 68.19	Text + Layout + Image
$\text{LayoutLMv3}_{\text{BASE}}$	92.08 → 67.69	Text + Layout + Image

TABLE I: Variations in F1 score after the FUNSD document layout adjustments.

The contributions of this paper are summarized as follows:

- We propose Uniform XY Cut, the algorithm for generating diverse permutations of bounding box arrangements to provide more accurate layout information and data augmentation for VrDU models.
- We propose the Visual-Graph Tagger, a module for

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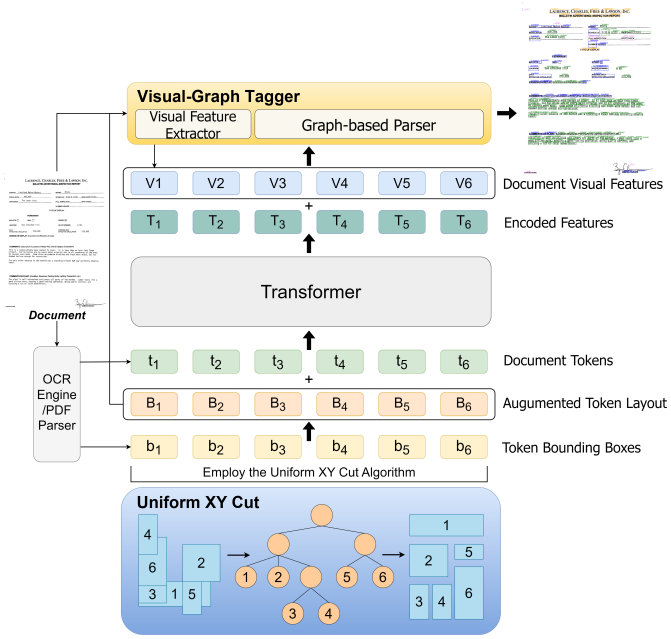


Fig. 2: An overview of RobustLayoutLM.

infusing visual and graph features into VrDU models without the need to modify the model architecture or pre-train a new model from scratch.

- We transfer our method to other VrDU models and achieve similar performance enhancement, demonstrating the effectiveness and generalizability of our approach.
- The improved model we propose, RobustLayoutLM, exhibits better performance on public visual document datasets with different layout conditions.

II. RELATED WORK

A. Visually-Rich Document Understanding

Visual document understanding has gone through several developmental stages: rule-based methods [1]–[5], machine learning methods [6]–[8], and deep learning methods [9]–[12]. Recently, with the widespread application of transformer-based models in the fields of natural language processing and computer vision, pre-training approaches [13]–[26] have become the mainstream for VrDU. Although they achieve notable gains, their performance significantly declines when the layout information changes from the original document layout. However, there has been little focus on the robustness of these VrDU models.

B. Taggers for Document Key Information Extraction

The BIO tagger and the BIES tagger are the most traditional and commonly utilized taggers for key information extraction, annotating a label for each token to locate entity positions in sequences. They do not provide any additional information for model output logits, just a common tagging format for information extraction. BROS [14] modifies the SPADE [27] to make it suitable for own pre-trained models. Our proposed Visual-Graph Tagger is composed of a visual feature extractor and a graph-based parser [14]. And we focus more on mitigating the performance loss caused by bias in document layout information, thus enhancing the robustness of VrDU models.

III. ROBUSTLAYOUTLM

The overall architecture of RobustLayoutLM is illustrated in Figure 2.

A. Uniform XY Cut Algorithm

The correct layout information is crucial for visual document understanding. In the presence of disordered bounding box sequences, our first step is to reconstruct the order of the bounding boxes to obtain the correct layout information. Given a set $B = \{b_1, b_2, \cdots, b_n\}$ of n document token bounding box, for each b_i within B, the vertical mapping $V_{b_i}(x)$ on the X-axis is defined by the width function:

$$V_{b_i}(x) = \begin{cases} 1 & \text{if } x^i_{min} \leq x \leq x^i_{max} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

And the horizontal project profile $H_{b_i}(y)$ of b_i is formulated as the height function:

$$H_{b_i}(y) = \begin{cases} 1 & \text{if } y^i_{min} \leq y \leq y^i_{max} \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

For the token bounding box set B, the value of vertical and horizontal projection profile correspondingly represent the number of bounding box that been projected onto the X-axis and Y-axis. Define the minimum and maximum values of the set B on the X-axis as x_{min} and x_{max}. By summing up the vertical projection profile on the X-axis for each individual bounding box, we define the vertical projection profile for the set B:

$$V_B = \sum_{i=1}^{N} V_{b_i}(x) \tag{3}$$

where $x \in \mathbb{Z}^{[x_{\min}, x_{\max}]}$. The minimum and maximum values of box set B in the horizontal direction are designated as y_{min} and y_{max}. By adopting the similar approach, the horizontal projection profile is formulated with $y \in \mathbb{Z}^{[y_{\min}, y_{\max}]}$:

$$H_B = \sum_{i=1}^{N} H_{b_i}(y) \tag{4}$$

As shown in Algorithm 1, the Uniform XY Cut algorithm takes a set of bounding boxes B, a parameter θ, and a threshold λ as inputs. Before performing horizontal or vertical projection on the bounding box set, we introduce a brand new augmentation strategy to preprocess the layout information. For each bounding box in the set B, we generate a random value k from $U(0, 1)$. If $|k| > \lambda$, shifting the bounding box $k \cdot \theta$ pixels along both the X-axis and the Y-axis. λ serves as the threshold to determine whether the bounding box should be shifted. In this paper, we set $\lambda = 0.5, \theta = 5$

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Method	F1	Params	Pre-training Modality
$\text{BERT}_{\text{BASE}}$	60.26	110M	Text
$\text{RoBERTa}_{\text{BASE}}$	66.48	125M	Text
$\text{BROS}_{\text{BASE}}$	83.05	110M	Text + Layout
$\text{LayoutLM}_{\text{BASE}}$	78.66	113M	Text + Layout
$\text{DocFormer}_{\text{BASE}}$	83.34	183M	Text + Layout + Image
$\text{SelfDoc}_{\text{BASE}}$	83.36	137M	Text + Layout + Image
$\text{XYLayoutLM}_{\text{BASE}}$	83.35	-	Text + Layout + Image
$\text{LayoutLMv2}_{\text{BASE}}$	82.76	200M	Text + Layout + Image
$\text{LiLT}_{\text{BASE}}$	88.41	130M	Text + Layout
$\text{BERT}_{\text{LARGE}}$	65.63	340M	Text
$\text{RoBERTa}_{\text{LARGE}}$	70.72	340M	Text
$\text{LayoutLM}_{\text{LARGE}}$	77.89	343M	Text + Layout
$\text{BROS}_{\text{LARGE}}$	84.52	340M	Text + Layout
$\text{StructuralLM}_{\text{LARGE}}$	85.14	355M	Text + Layout
$\text{LayoutLMv2}_{\text{LARGE}}$	84.20	426M	Text + Layout + Image
OURS	**88.70**	160M	Text + Layout

TABLE II: Performance comparisons on the FUNSD dataset. **Bold** indicates the best performance.

as the default hyper-parameters for the Uniform XY Cut algorithm based on the experimental results. Upon obtaining the augmented bounding box set B_{aug} from the given layout information, we calculate the projection profiles in either horizontal or vertical direction. New child nodes are obtained by dividing the valleys of the projection profiles where the width exceeds the predefined threshold value. In this way obtaining the correct order of the set B_{aug} is divided into subtasks corresponding to the number of child nodes. The recursive horizontal and vertical divisions are repeated until there are not enough number of valleys in each child node for further division. In the end, we reorder the set B_{aug} according to the indices on the leaf nodes. After implementing the Uniform XY Cut algorithm on the token bounding box set B, we get the augmented set with improved sequence order. Our augmentation strategy differs from [19] in two main aspects:

- In our augmentation strategy, the bounding boxes are shifted as a whole, ensuring that shapes of original text blocks remain unchanged during the movement, thereby maintaining data consistency.
- Our method determines whether to generate data augmentation by shifting the bounding box based on a uniform distribution, not a normal distribution. This choice is motivated by the fact that improperly aligned bounding boxes have already introduced a higher level of uncertainty, and we aim to keep the augmentation proportion within a reasonable limit under the same threshold.

To better illustrate how the recursive XY Cut is performed

Algorithm 1 Uniform XY Cut algorithm

Input: bounding box set B, hyper-parameters: θ, λ
Output: reordered bounding box set B_{ro}

1: Implement the augmentation strategy on the bounding box set B with the parameter θ and the threshold λ to get the augmented box set B_{aug}.
2: Perform horizontal or vertical projection on the bounding box set B_{aug}.
3: Search for the valleys in the horizontal projection profiles or vertical projection profiles.
4: Alternate the horizontal and vertical division at the valleys.
5: Recursively apply the step 3 and step 4 until no further divisions are attainable.
6: Collect all the leaf nodes as the output set B_{ro}.

on the augmented bounding box set, we give an example as shown at the bottom of Figure 2. Firstly, we vertically project the layout set to obtain the vertical projection profiles. Within this projection profile, two valleys are identified, resulting in two child nodes ($\{1, 2, 3, 4\}$ and $\{5, 6\}$). Then, we perform horizontal projection on the Y-axis for each of the two child nodes. The horizontal projection profiles of the first child node reveal two valleys, dividing the node into three sub-nodes ($\{1\}$, $\{2\}$ and $\{3, 4\}$). The first sub-node and the second sub-node each have only one element in them that can no longer be recursively cut and thus become leaf nodes. The third sub-node contains more than one element, therefore needs to be further divided. The recursive xycut algorithm is iteratively applied to the nodes in the last layer of the XY Tree until all leaf nodes contain only one element. The element within the leaf nodes corresponds to the final arrangement indices of the layout set. The detailed pseudocode can be found in the Appendix.

B. Visual-Graph Tagger

A visual document can provide four types of modal information: textual information, layout information, visual information and graph information between tokens. When errors in document parsing by the OCR engine/PDF parser lead to layout inaccuracies, it is intuitive to introduce visual and graph modal information that is not affected by the mis-layout to enrich the feature representation, thus reduce the impact of erroneous layout information on the model performance. However, numerous existing VrDU models are pretrained based on textual and layout modalities, and the most commonly used BIO and BIES taggers are merely tagging formats, which do not provide any additional information for document feature representation. To this end, we propose the Visual-Graph Tagger, which enables the integration of visual and graph features into the text-and-layout VrDU models through the tagger, without the need for an additional round of model pre-training or modifying the model structure.

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The Visual-Graph Tagger consists of two parts: a visual feature extractor and a graph-based parser. The visual feature extractor use ResNet-101 [28] as the visual backbone network to generate the image feature map of the document image I. For each token in every text block within the document, there is a corresponding bounding box indicating its position in the document, so the corresponding feature map for each token can be obtained using RoIAlign [29]. For a text sequence $S = \{t_0, t_1, \cdots, t_N\}$ with the length of N, the representation $\widehat{S} = \left\{\widehat{t_0}, \widehat{t_1}, \cdots, \widehat{t_N}\right\}$ output from the final layer of transformer, thus the layout-text-image contextual representation $\mathbf{X}$ that incorporates additional visual features can be obtained:

$$\mathbf{X} = \text{Proj}(\text{Flatten}(\text{ROI_Align}(\text{ResNet}(I)))) + \widehat{S} \quad (5)$$

The graph-based parser builds a directed graph at the token level to capture the relationships between multiple tokens within the same text block. And it breaks down the document information extraction task into two subtasks: the first token classification task and the subsequent token link task.

As shown in Figure 3(a), the first token classification task aim to identify the label to which the text block or the sequence belongs.

$$p_{\text{fc}}\ (\mathbf{x}_i) = \text{softmax}\left(\boldsymbol{W}^{\text{fc}}\ \mathbf{x}_i\right) \quad (6)$$

where $\boldsymbol{W}^{\text{fc}} \in \mathbb{R}^{(n+1)\times H}$ is a linear transition matrix, n is the number of target classes, and $\mathbf{x}_i$ denotes the layout-text-image contextual representation of i_{th} token.

Tokens within the same text block belong to the same category, therefore it is necessary to determine which text tokens are contained within the same text block. As depicted in Figure 3(b) The subsequent token link task is able to distinguish all the text tokens in the same block or sentence and link them together:

$$\begin{aligned} \mathbf{X}_{\text{sl}} &= \boldsymbol{W}^{\text{sl}_1}\left[\mathbf{x}_1; \mathbf{x}_2; \ldots; \mathbf{x}_N; \mathbf{x}_{\text{sl}}\right], \\ p_{\text{sl}}\ (\mathbf{x}_i) &= \text{softmax}\left(\left(\boldsymbol{W}^{\text{sl}_2}\mathbf{x}_i\right)^{\top}\ \mathbf{X}_{\text{sl}}\right)^{\top} \end{aligned} \quad (7)$$

where $\mathbf{x}_{\text{sl}}$ signifies the end of the sequence.

After determining the label of the text block through the first token classification and then connecting the tokens in the same text block through the subsequent token link, the Visual-Graph Tagger can identify the label corresponding to each token in the document.

IV. EXPERIMENTS AND RESULTS

A. Experiment Settings

Datasets. Our experiments are conducted base on the public datasets FUNSD [30] and XFUND [18]. The FUNSD is a noisy scanned document understanding dataset sampled from the RVL-CDIP dataset [31], which is sourced from a variety of forms, including invoices, receipts, and application forms. This diversity makes FUNSD an excellent benchmark for evaluating the robustness of a model under different layout conditions. The XFUND dataset is a multilingual document understanding dataset that includes seven languages: Chinese, Japanese, Spanish, French, Italian, German and Portuguese. We simulate the case of incorrect layout information by shuffling the bounding boxes order in FUNSD and XFUND dataset with a fixed random seed, generating the datasets s-FUNSD and s-XFUND. For the FUNSD dataset, the token bounding boxes are reordered according to a heuristic rule to obtain the datasets xy-FUNSD, x-FUNSD, and y-FUNSD, where the bounding boxes of text tokens are reorganized in ascending order along both the X-axis and the Y-axis, ascending X-axis and ascending Y-axis, respectively.

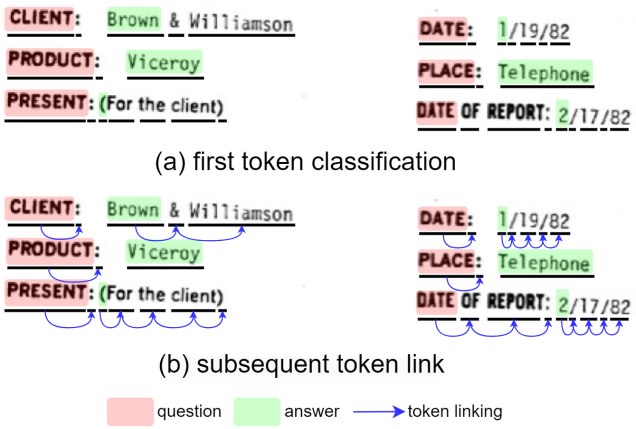


Fig. 3: Visual depictions of the graph-based parser.

Training details. Since the text-layout backbone LiLT needs to be combined with the different pre-trained textual models for performing multilingual understanding, we trained two models RobustLayoutLM and RobustLayoutXLM, targeting English document understanding and non-English document understanding. We initialized their weights with LiLT[EN-R]$_{\text{BASE}}$ [21] and LiLT[Info-XLM]$_{\text{BASE}}$ [21] separately and as their baselines. On the FUNSD dataset, the batch size size is set to 8, employ AdamW [32] optimizer with a learning rate of 5e-5, and train for 100 epochs. While in the experiments on the XUNSD dataset, the batch size is set to 4.

B. Experiment Results

With Default Layout Information. We compare our method with text based methods [33], [34], layout-text based methods [15], [21], [24] and visual-layout-text based methods [16], [22], [23] on the FUNSD dataset, with the results presented in Table II. Our method RobustLayoutLM achieves the highest F1 score among the listed methods. Against the layout-text model StructuralLM with large parameter size, our method uses fewer than half the parameters and still achieves a 4.18% increase in F1 score. Even in the absence of image features during the pre-training stage, our method demonstrates superior performance with fewer parameters relative to the model LayoutLMv2(200M→ 160M, 82.76 →

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Method	Pre-training Modality	ZH	JA	ES	FR	IT	DE	PT	Avg.
LayoutXLM	Text + Layout + Image	89.24	79.21	75.50	79.02	80.82	82.22	79.03	80.56
XYLayoutLM	Text + Layout + Image	**91.76**	**80.57**	76.87	79.97	81.75	83.35	80.01	82.04
$\text{LiLT}_{\text{BASE}}$	Text + Layout	89.38	79.64	**79.11**	79.53	83.76	82.31	82.20	82.28
OURS	Text + Layout	89.09	78.86	79.06	**80.39**	**84.71**	**84.98**	**82.47**	**82.79**

TABLE III: Performance comparisons on the XFUND dataset.

Method	F1	Params	Pre-training Modality
$\text{BERT}_{\text{BASE}}$	60.26	110M	Text
$\text{RoBERTa}_{\text{BASE}}$	66.48	125M	Text
$\text{BROS}_{\text{BASE}}$	61.04	110M	Text + Layout
$\text{LayoutLM}_{\text{BASE}}$	66.62	113M	Text + Layout
$\text{LayoutLMv2}_{\text{BASE}}$	68.19	200M	Text + Layout + Image
$\text{LiLT}_{\text{BASE}}$	68.93	130M	Text + Layout
$\text{LayoutLMv3}_{\text{BASE}}$	67.69	133M	Text + Layout + Image
$\text{BERT}_{\text{LARGE}}$	65.63	340M	Text
$\text{LayoutLM}_{\text{LARGE}}$	67.93	343M	Text + Layout
$\text{BROS}_{\text{LARGE}}$	45.38	340M	Text + Layout
OURS	**72.02**	160M	Text + Layout

TABLE IV: Performance comparisons on the s-FUNSD.

87.85), which utilizes visual information in its pre-training process.

Nearly 40% of digital documents on the web are in non-English languages [18], which creates a huge need for multilingual document understanding. To evaluate the performance of RobustLayoutXLM in multilingual document understanding, we compare our model with other available multilingual VrDU models on the XFUN dataset. As shown in Table III, the result shows that our method RobustLayoutXLM achieves the highest average F1 score across seven languages, with an improvement of 0.62% compared to the baseline. On the French, Italian, German and Portuguese subsets, F1 scores have shown improvement over the baseline, with the largest gain seen in the German subset, where the F1 score improved by 3.24%. But in the other subsets, the F1 score is lower than the baseline, we conjecture that the variance in the model performance across different language subsets can be attributed to the distinct document formats and layouts across different countries, lead to variations in the additional features captured by our method.

With Different Layout Information. Document understanding tasks rely on OCR tools to obtain layout information during downstream inference, Different OCR tools may produce different layout information due to distinct recognition order. To validate the robustness of RobustLayoutLM and RobustLayoutXLM under multiple layout conditions, in addition to the experiments on the default layout information of the FUNSD and XFUND datasets, we have also performed experiments on the derivatives of these datasets with different layout information. As shown in Table IV, for BERT and RoBERTa, since only text is used for pre-training, the disorder of layout information does not affect its performance. In comparison with LayoutLM, BROS, and LiLT, which utilize both text information and layout information during the pre-training stage, our model exhibits superior performance to all these models. Compared to $\text{LayoutLMv3}_{\text{BASE}}$, our model even pre-training without image feature, however, still earning better performance (67.69 $\rightarrow$ 72.02). For the large model size, due to the fact that these models only rely on textual information and layout information, any errors in the layout information can significantly impact the performance of the model. Our model outperforms them employing no more than half of the parameter volume, gaining 58.70% (45.38 $\rightarrow$ 72.02) F1 score boost with only 47.06% (340M vs 160M) of the parameter volumes compared to $\text{BROS}_{\text{LARGE}}$.

The results from adding heuristic rules to document layout are depicted in Table VI. Among all the methods, BERT shows the worst performance. It is noteworthy that the BROS model, which undergoes pre-training with both layout and textual features, performs even worse than RoBERTa, which solely relies on text features. This finding validates our viewpoint that incorrect layout information can limit the expression of other modal information. Moreover, we observe that when there are errors in the document layout information, the performance gains from using models with larger parameter sizes are quite limited. $\text{LayoutLM}_{\text{LARGE}}$, with a parameter count threefold that of $\text{LayoutLM}_{\text{BASE}}$, only offers an average F1 score increase of 0.05% across these datasets. Moreover, the larger parameter-sized $\text{LayoutLM}_{\text{LARGE}}$ shows performance degradation (67.40 $\rightarrow$ 67.01) compared to $\text{LayoutLM}_{\text{BASE}}$ on the xy-FUNSD dataset. Our model continues to achieve the highest F1 scores across heuristic rules ordered datasets.

In order to validate the robustness of our method across multiple languages, we compare RobustLayoutXLM with public multilingual VrDU models on the s-XFUND. As shown in Table V, the result shows that our model outperforms all other methods across seven non-English languages, with the F1 scores on the seven language subsets improve by an average of 5.50% (67.41 $\rightarrow$ 71.12) and 5.35% (67.51 $\rightarrow$ 71.12) compared to the baseline and LayoutXLM, respectively. It is noteworthy that our model exhibits the smallest decline in average F1 score across seven languages when confronted with layout errors, with a particularly minimal

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Method	Pre-training Modality	ZH	JA	ES	FR	IT	DE	PT	Avg.
LayoutXLM	Text + Layout + Image	86.75	73.59	51.31	61.57	66.31	67.76	65.26	67.51
$\text{LiLT}_{\text{BASE}}$	Text + Layout	85.95	75.49	53.51	66.39	63.58	63.02	63.93	67.41
OURS	Text + Layout	**87.64**	**77.75**	**60.95**	**68.40**	**66.48**	**68.39**	**68.24**	**71.12**

TABLE V: Performance comparisons on the s-XFUND dataset.

Method	Pre-training Modality	Params	xy-FUNSD F1	x-FUNSD F1	y-FUNSD F1
$\text{BERT}_{\text{BASE}}$	Text	110M	60.26	60.26	60.26
$\text{RoBERTa}_{\text{BASE}}$	Text	125M	66.48	66.48	66.48
$\text{BROS}_{\text{BASE}}$	Text + Layout	110M	61.04	61.98	62.71
$\text{LayoutLM}_{\text{BASE}}$	Text + Layout	113M	67.40	66.37	66.84
$\text{LayoutLMv2}_{\text{BASE}}$	Text + Layout + Image	200M	69.72	67.99	69.98
$\text{LiLT}_{\text{BASE}}$	Text + Layout	110M	70.04	69.80	69.17
$\text{LayoutLMv3}_{\text{BASE}}$	Text + Layout + Image	133M	67.37	69.43	69.28
$\text{BERT}_{\text{LARGE}}$	Text	340M	65.63	65.63	65.63
$\text{LayoutLM}_{\text{LARGE}}$	Text + Layout	343M	67.01	66.51	67.20
$\text{BROS}_{\text{LARGE}}$	Text + Layout	340M	63.69	64.44	62.71
OURS	Text + Layout	160M	**71.14**	**70.65**	**70.56**

TABLE VI: Performance comparisons on the heuristic sorting datasets.

Method	F1
LayoutLM	66.62
+Visual-Graph Tagger	68.17
+Uniform XY Cut	68.36
+Uniform XY Cut&Visual-Graph Tagger	**69.38**

TABLE VII: Performance improvements on the LayoutLM.

Method	F1
Remove Uniform XY Cut	70.78
Uniform XY Cut (0.5,1)	70.82
Uniform XY Cut (0.3,5)	71.10
Uniform XY Cut (0.5,5)	**72.02**
Uniform XY Cut (0.7,5)	70.86
Uniform XY Cut (0.5,10)	71.53

TABLE VIII: F1 scores of RobustLayoutLM based on different combinations of λ and θ.

decline of only 1.65% and 1.43% on the Chinese and Japanese subsets.

From the above experiments, we can see that our method gains competitive results on standard VrDU benchmarks. Moreover, in the experiments on the non-original layout datasets, RobustLM and RobustXLM achieve the best performance in all of the different layout scenarios, which fully validating their robustness under multiple layout conditions.

Method	F1
LiLT	68.93
+Visual-Graph Tagger	70.78
+XY Cut	70.63
+Uniform XY Cut	70.07
+XY Cut&Visual-Graph Tagger	71.03
+Uniform XY Cut&Visual-Graph Tagger	**72.02**

TABLE IX: Ablation study of RobustLayoutLM.

C. Ablation Study

RobustLayoutXLM is the multilingual version of RobustLayoutLM, both of them adopt the same methods and model structure. We choose to conduct ablation experiments on RobustLayoutLM. As shown in Table IX, when the BIO tagger is replaced with the Visual-Graph Tagger, there is a 2.68% increase in the F1 score, indicating that adding extra visual and graph information to the model proves effective. The use of the Uniform XY Cut algorithm to reorder incorrect token bounding boxes brings a 1.65% increase in the F1 score. Compared to traditional XY Cut, the standalone use of the Uniform XY Cut does not result in a larger improvement. We speculate this is because the layout information generated by the Uniform XY Cut algorithm is more varied compared to that of the traditional XY Cut algorithm, making it challenging for the model to fully utilize the diverse layout information. However, after introducing

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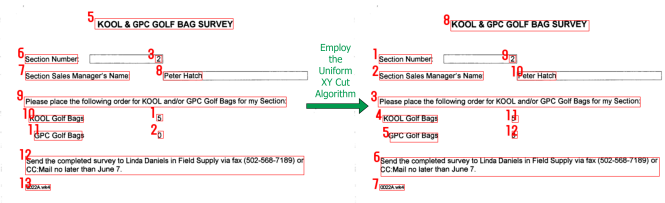


Fig. 4: Visualizations of the Uniform XY Cut.

the Visual-Graph Tagger, as shown in the fifth and sixth rows, the combination of the Visual-Graph Tagger and the Uniform XY Cut achieves a 1.39% higher F1 score compared to that of the combination with traditional XY Cut. The diverse layout information generated by the Uniform XY Cut algorithm can provide richer layout information for the multimodal integration within the Visual-Graph Tagger, leading to the better performance than when combined with the traditional XY Cut. From the last row, it can be seen that when the two components are employed jointly, the F1 score improvement is greater than the sum of the F1 improvement of the two individual components, which achieves 4.48% F1 score improvement compared to the baseline model.

To validate the generalizability of our method on other text-layout backbone networks, we port our method to another backbone, LayoutLM, as shown in Table VII. It can be observed that the inclusion of both the Uniform XY Cut and the Visual-Graph Tagger are able to increase the F1 score of LayoutLM, demonstrating the versatility of our approach.

As shown in Table VIII, we set different combinations of λ and θ to explore the impact of varying λ and θ within the Uniform XY Cut on model performance. There is a drop in the F1 score when the Uniform XY Cut algorithm is removed. As shown in the subsequent five rows, the introduce of the Uniform XY Cut results in performance improvements at various levels. Among these, the combination of $\lambda = 0.5$ and $\theta = 5$ leads to the highest F1 score compared to other λ, θ pairings, we set it as the default hyper-parameters for the Uniform XY Cut algorithm.

D. Visualizations

We visualize the ordering of token bounding boxes in Figure 4. It can be seen that our proposed Uniform XY Cut algorithm successfully reorders the bounding boxes in a further optimized manner.

CONCLUSION

In this paper, we present the novel Uniform XY Cut algorithm and the Visual-Graph Tagger module, where the augmented layout information generated by the Uniform XY Cut is beneficial for the multimodal feature representation within the Visual-Graph Tagger. The enhanced feature representation can alleviate the dependence of layout-and-text based VrDU methods on a single layout information. Experiment results on seven VrDU datasets demonstrate the effectiveness and generalizability of our method.

In future research, we will further explore the the extraction of document layout and visual features to cope with visual documents with more complex layouts. Additionally, we will investigate multiple fusion methods of document multimodal features to enhance the model's multilingual capabilities.

Algorithm 2 Uniform XY Cut algorithm

Input: bounding box set: B, initial direction: dir, hyper-parameters: θ, λ
Output: reordered bounding box set: B_{ro}

function CUT($bboxes$, $direction$)
  $res \leftarrow [\,]$
  $c_i \leftarrow 0$
  $s_i, s_box \leftarrow$ **sort** $bboxes$ by $direction$
  $n_dir \leftarrow$ **shift** to the other direction.
  for $i \leftarrow 0$ **to** $len(s_box) - 1$ **do**
    if $direction$ is X-axis **then**
      if $s_box[i][2] < s_box[i+1][0]$ **then**
        $t_res \leftarrow \text{CUT}(s_box[c_i : i+1], n_dir)$
        $res \leftarrow res + [s_i[c_i + j]$ for j in $t_res]$
        $c_i \leftarrow i + 1$
      end if
    else
      if $s_box[i][3] < s_box[i+1][1]$ **then**
        $t_res \leftarrow \text{CUT}(s_box[c_i : i+1], n_dir)$
        $res \leftarrow res + [s_i[c_i + j]$ for j in $t_res]$
        $c_i \leftarrow i + 1$
      end if
    end if
  end for
  $res \leftarrow res + s_i[c_i : len(s_box)]$
  return res
end function
function UNIXYCUT($bboxes, direction, theta, lambda$)
  for b in $bboxes$ **do**
    $k \leftarrow$ Random Init from $U(0, 1)$
    if $|k| \geq lambda$ **then**
      $b[0] += k \cdot theta, b[1] += k \cdot theta$
      $b[2] += k \cdot theta, b[3] += k \cdot theta$
    end if
  end for
  $res_indexs \leftarrow \text{CUT}(bboxes, direction)$
  $res_bboxes \leftarrow [bboxes[i]$ for i in $res_indexs]$
  return res_bboxes
end function
$B_{ro} \leftarrow \text{UNIXYCUT}(B, dir, \theta, \lambda)$

APPENDIX

The detailed pseudocode of the Uniform XY Cut algorithm is shown in Algorithm 2.

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